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LINUX FUNDAMENTALS CA-2 PROJECT

SUBMITTED BY: RISHIKESH

REG NO: 12310772

Project:

Logical Volume Management with Disk Expansion & ACL Configuration

Chronological Explanation of Steps:

1. Set registration number in the shell prompt -> export PS1="[12307284 \W]\\\$ "
2. Use Host Key to take screenshots generally right ctrl+e.
3. Check Drive Status -> lsblk
4. Powering VM off to add virtual drives i.e (1gb,3gb,4gb)

```
[sid@vbox ~]$ export PS1="[12310772 \W]\\$ "
[12310772 ~]$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0   20G  0 disk
├─sda1                              8:1      0    1G  0 part /boot
├─sda2                              8:2      0   19G  0 part
│   └─rhel_vbox-root                253:0    0   17G  0 lvm  /
│       └─rhel_vbox-swap            253:1    0    2G  0 lvm  [SWAP]
sdb                                  8:16     0    1G  0 disk
sdc                                  8:32     0    3G  0 disk
sdd                                  8:48     0    4G  0 disk
sr0                                  11:0     1  58.5M  0 rom  /run/media/sid/VBox_GAs_7.1.10
[12310772 ~]$ sudo pvcreate /dev/sdb /dev/sdc /dev/sdd
[sudo] password for sid:
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
Physical volume "/dev/sdd" successfully created.
[12310772 ~]$ sudo vgcreate projectvg /dev/sdb /dev/sdc /dev/sdd
Volume group "projectvg" successfully created
[12310772 ~]$ sudo lvcreate -L 5G -n projectlv projectvg
WARNING: ext4 signature detected on /dev/projectvg/projectlv at offset 1080. Wipe it? [y/n]: y
Wiping ext4 signature on /dev/projectvg/projectlv.
Logical volume "projectlv" created.
[12310772 ~]$ sudo mkfs.ext4 /dev/projectvg/projectlv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 06f92dac-5ca6-4a1d-9bbd-f7a8b6f926f4
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[12310772 ~]$
```

5. Create three physical volumes (using the three added disks) -> `sudo pvcreate /dev/sdb /dev/sdc /dev/sdd`
6. Create a volume group named projectvg -> `sudo vgcreate projectvg /dev/sdb /dev/sdc /dev/sdd`
7. Create a logical volume of 5GB -> `sudo lvcreate -L 5G -n projectlv projectvg`
8. Format the logical volume with ext4 filesystem -> `sudo mkfs.ext4 /dev/projectvg/projectlv`
9. Create a mount point directory -> `sudo mkdir /mnt/projectdata`
10. Mount the logical volume to the directory -> `sudo mount /dev/projectvg/projectlv /mnt/projectdata`
11. Make the mount permanent after rebootsudo nano /etc/fstab
10. Create a user for ACL configuration -> `sudo useradd projectuser -> sudo passwd projectuser`
`sudo setfacl -m u:projectuser:rwx /mnt/projectdata`
13. Verify ACL permissions -> `getfacl /mnt/projectdata`
14. Resize logical volume by 2GB to make it 7GB -> `sudo lvextend -r -L +2G /dev/projectvg/projectlv`
15. `sudo resize2fs /dev/projectvg/projectlv`(if you want to use remaining 2 gb)

	Virtual Size	Actual Size
Attached		
Redux.vdi	20.00 GB	5.47 GB
Not Attached		
Redux_1.vdi	1.00 GB	2.00 MB
Redux_2.vdi	3.00 GB	2.00 MB
Redux_3.vdi	4.00 GB	2.00 MB

```
#
# /etc/fstab
# Created by anaconda on Fri Jul 4 05:52:54 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rhel_vbox-root / xfs defaults 0 0
UUID=aa77ffa9-84ff-48f7-9837-cfeb186789be /boot xfs defaults 0 0
/dev/mapper/rhel_vbox-swap none swap defaults 0 0
/dev/projectvg/projectlv /mnt/projectdata ext4 deafaults,acl 0 0
```

```
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 2587564a-e240-4202-809c-a8f9e53d445a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736
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Allocating group tables: done
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```
[12310772 ~]$sudo mount /dev/projectvg/projectlv /mnt/projectdata
```

```
[12310772 ~]$df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	886M	0	886M	0%	/dev/shm
tmpfs	355M	5.7M	349M	2%	/run
/dev/mapper/rhel_vbox-root	17G	4.5G	13G	27%	/
/dev/sda1	960M	365M	596M	38%	/boot
tmpfs	178M	96K	178M	1%	/run/user/1000
/dev/sr0	59M	59M	0	100%	/run/media/sid/VBox_GAs_7.1.10
/dev/mapper/projectvg-projectlv	4.9G	24K	4.6G	1%	/mnt/projectdata

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[12310772 ~]$sudo setfacl -m u:sid:rw /mnt/projectdata
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setfacl: Option -m: Invalid argument near character 3
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```
[12310772 ~]$getfacl /mnt/projectdata
```

```
getfacl: Removing leading '/' from absolute path names
```

```
# file: mnt/projectdata
```

```
# owner: root
```

```
# group: root
```

```
user::rw
```

```
user:sid:rw
```

```
group::r-x
```

```
mask::rw
```

```
other::r-x
```

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mask::rw
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```
other::r-x
```

```
[12310772 ~]$
```


Proofs ACL/Resizing/LVM

```
Cur LV                2
Open LV                2
Max PV                 0
Cur PV                1
Act PV                 1
VG Size                <19.00 GiB
PE Size                4.00 MiB
Total PE              4863
Alloc PE / Size       4863 / <19.00 GiB
Free PE / Size        0 / 0
VG UUID                S3Nv3x-9ufP-q3Z0-wE4p-WfXB-dscj-4mod6P

--- Volume group ---
VG Name                projectvg
System ID
Format                lvm2
Metadata Areas         3
Metadata Sequence No  1
VG Access              read/write
VG Status              resizable
MAX LV                0
Cur LV                0
Open LV                0
Max PV                0
Cur PV                3
Act PV                3
VG Size                <7.99 GiB
PE Size                4.00 MiB
Total PE              2045
Alloc PE / Size       0 / 0
Free PE / Size        2045 / <7.99 GiB
VG UUID                jgFD1H-u3ge-NhML-59JN-4qzs-vMkb-0MdWdw
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Filesystem            Size  Used Avail Use% Mounted on
devtmpfs              4.0M   0  4.0M   0% /dev
tmpfs                 886M   0  886M   0% /dev/shm
tmpfs                 355M  5.7M  349M   2% /run
/dev/mapper/rhel_vbox-root 17G  4.5G  13G  27% /
/dev/sda1             960M  365M  596M  38% /boot
tmpfs                 178M   96K  178M   1% /run/user/1000
/dev/sr0              59M   59M    0 100% /run/media/sid/VBox_GAs_7.1.10
/dev/mapper/projectvg-projectlv 4.9G   24K  4.6G   1% /mnt/projectdata
[12310772 ~]$sudo lvextend -L 7G /dev/projectvg/projectlv
Size of logical volume projectvg/projectlv changed from 5.00 GiB (1280 extents) to 7.00 GiB (1792 extents).
Logical volume projectvg/projectlv successfully resized.
[12310772 ~]$
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	20G	0	disk	
sda1	8:1	0	1G	0	part	/boot
sda2	8:2	0	19G	0	part	
rhel_vbox-root	253:0	0	17G	0	lvm	/
rhel_vbox-swap	253:1	0	2G	0	lvm	[SWAP]
sr0	11:0	1	58.5M	0	rom	/run/media/sid/VBox_GAs_7.1.10